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Performance Analysis of Lump Sum and Installment Investing Across Mutual Fund Categories

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Abstract

Indian retail investors generally choose between two ways into equity mutual funds: contributing through a Systematic Investment Plan (SIP) or introducing capital as a single lump sum. Prior work has generally leaned toward SIPs, pointing to smoother entry pricing and steadier accumulation over time. Yet two limitations persist across this body of work. Comparisons of SIP and lump-sum outcomes are usually conducted without separating funds by market-capitalization tier, even though large, mid and small-cap schemes behave very differently in terms of volatility and growth potential. Meanwhile, studies that separates the funds by tier tend to depend on a single risk perspective, most often plain volatility or one reward-to-risk ratio, leaving out how each method performs once downside movement or benchmark-relative skill is accounted for separately. This project sets out to close both gaps together. Drawing on published NAV records for a set of large, mid and small-cap Indian equity schemes spanning 2015 to 2025. A stretch covering four distinct downturns and recoveries. The analysis will compute standard deviation alongside the Sharpe, Sortino and Treynor measures plus Jensen's Alpha for both investment routes within each tier. Doing so separates total variability, downside-specific risk, market-linked risk and manager-driven excess return into distinct pictures rather than one blended verdict. The intent is to establish whether SIP's edge holds across every tier or narrows to particular segments, giving advisors and fund houses a more precise, tier-specific basis for guiding this choice.

Keywords: *Mutual Funds; SIP; Lump-sum Investment; Risk-Adjusted Performance; Sharpe Ratio; Sortino Ratio; Treynor Ratio; Jensen's Alpha; Large-cap; Mid-cap; Small-cap; India*

2. Introduction

2.1 Background of the Study

Mutual funds have become the principal vehicle through which Indian households participate in equity markets. Industry assets under management stood at approximately ₹87.08 lakh crore in August 2026, with monthly Systematic Investment Plan (SIP) contributions of ₹32,297 crore flowing from more than ten crore contributing SIP accounts, and SIP assets alone accounting for close to 21 per cent of total industry assets (Association of Mutual Funds in India [AMFI], 2026). The scale of this shift is comparatively recent: monthly SIP collections have risen roughly tenfold over the past decade, and the proportion of SIP assets held for more than five years has more than doubled since 2021. The SIP has consequently moved from being one product feature among many to being the default architecture of retail equity participation in India.

A SIP commits a fixed rupee amount at regular intervals, so the number of units purchased rises when the Net Asset Value (NAV) falls and falls when the NAV rises. Internationally this mechanism is studied as dollar-cost averaging (DCA). The alternative route, a lump-sum investment, places the entire corpus in the market at a single point in time and leaves it invested for the full holding period. The two routes therefore differ not in the securities held but in the timing and path by which capital enters those securities - a distinction that matters precisely because equity markets are volatile and their volatility is not uniform across market-capitalisation segments.

2.2 Context of the Research Problem

The choice between the two routes is made continuously by investors and advisers, and it is made with surprisingly thin evidence. Theoretical finance has long held that staggering entry is suboptimal in expected-return terms whenever the equity risk premium is positive, because capital held back earns the lower risk-free rate for part of the horizon (Constantinides, 1979; Rozeff, 1994). Practitioner and Indian academic literature, by contrast, reports repeatedly that SIPs deliver better outcomes, particularly in volatile markets (Dwivedi & Bhadouria, 2025; Parulekar, 2025). This apparent contradiction is partly definitional, because the two literatures compare different things, and partly evidential, because the empirical Indian work rarely controls

for the characteristics of the funds being compared or tests whether the observed differences are statistically distinguishable from zero.

Two controls are conspicuously absent. The first is segmentation by market capitalisation. Large-, mid- and small-cap schemes differ systematically in volatility, in the depth of their drawdowns and in the length of their recoveries; a route whose advantage derives from buying through declines should therefore not be expected to perform identically across the three tiers. The second is the choice of evaluation lens. Total volatility, downside volatility, systematic risk and benchmark-relative skill are conceptually distinct, and a comparison that reports only one of them cannot say whether an observed advantage reflects better compensation for risk borne or genuine outperformance of the benchmark.

2.3 Significance of the Study

Resolving these questions has direct practical value. With over ten crore active SIP accounts, even a small misattribution of where SIP's advantage comes from has consequences for how advisers allocate client cash flows and how asset management companies position their products. If SIP's advantage is a mechanical consequence of averaged entry, the appropriate recommendation is tied to an investor's cash-flow pattern and to the volatility of the segment being entered. If, instead, SIP were shown to improve benchmark-relative performance, the recommendation would be far broader. Establishing which of these is true, and whether it holds equally in every tier, is therefore a question of practical consequence rather than merely of academic interest.

2.4 Research Problem

The research problem addressed in this study is that the widely accepted conclusion that SIP investing outperforms lump-sum investing in Indian equity mutual funds has not been tested simultaneously across market-capitalisation tiers and across multiple, conceptually distinct risk-adjusted lenses, leaving it unclear whether the advantage is uniform across segments and whether it represents a timing effect or an improvement in benchmark-relative performance.

2.5 Research Questions

- RQ1: Over the period 2015–2025, does SIP investing deliver higher annualised and risk-adjusted returns than lump-sum investing across Indian large-cap, mid-cap and small-cap equity mutual funds?
- RQ2: Does the magnitude and the statistical significance of any SIP advantage vary across the three market-capitalisation tiers?
- RQ3: When performance is decomposed using the Capital Asset Pricing Model, is any SIP advantage attributable to benchmark-relative skill (Jensen's Alpha) or to the path by which capital enters the market?

2.6 Hypotheses

The following hypotheses are tested using paired-sample t-tests on scheme-level differences (SIP value minus lump-sum value), both pooled across the nine schemes and separately within each tier:

- $H_0 1$: There is no significant difference in annualised return between the SIP and lump-sum routes. $H_1 1$: SIP yields a significantly higher annualised return.
- $H_0 2$: There is no significant difference in the Sharpe ratio between the two routes. $H_1 2$: SIP yields a significantly higher Sharpe ratio.
- $H_0 3$: There is no significant difference in the Sortino ratio between the two routes. $H_1 3$: SIP yields a significantly higher Sortino ratio.
- $H_0 4$: There is no significant difference in the Treynor ratio between the two routes. $H_1 4$: SIP yields a significantly higher Treynor ratio.
- $H_0 5$: There is no significant difference in Jensen's Alpha between the two routes. $H_1 5$: Jensen's Alpha differs significantly between the two routes.

2.7 Overview of the Study

The study is quantitative, analytical and simulation-based, and relies entirely on published secondary data. Nine Direct-Growth equity schemes - one large-cap, one mid-cap and one small-cap fund from HDFC, ICICI Prudential and SBI - are tracked over 132 months from January 2015 to December 2025. For each scheme, two portfolios are constructed from the same

total capital of ₹13,20,000: one built from 132 monthly installments of ₹10,000 and one from a single commitment in January 2015. Both are marked to market monthly, and each is evaluated on annualised return, standard deviation, downside deviation, Beta, and the Sharpe, Sortino and Treynor ratios along with Jensen's Alpha, measured against the tier-appropriate Total Returns Index. Differences between the routes are then tested for statistical significance. The remainder of the report reviews the relevant literature, sets out the methodology in detail, presents the results, discusses their interpretation and concludes with limitations and directions for further work.

3. Literature Review

The literature bearing on this study falls into four connected strands: the theoretical case against staggered entry, the behavioural and measurement-based case in its defence, the Indian empirical evidence on SIP versus lump-sum investing, and the evaluation frameworks used to judge fund performance. Each strand is examined below for what it establishes, where its findings diverge and what it leaves unresolved.

3.1 The Theoretical Case Against Staggered Entry

The oldest and analytically firmest position in this literature is that staggering the entry of a fixed corpus is suboptimal. Constantinides (1979) demonstrated within a mean–variance framework that dollar-cost averaging is dominated by a flexible policy that conditions on available information, because a fixed contribution schedule set in advance ignores subsequent market information. Rozeff (1994) reached a compatible conclusion by a different route, showing that once outcomes are adjusted for risk, immediate full investment is superior provided the equity risk premium is positive, since capital withheld from the market earns only the risk-free rate. Williams and Bacon (1993) reported the empirical counterpart in United States data, and Vanguard's widely cited rolling-window study extended the finding to the United Kingdom and Australia, concluding that immediate investment outperformed twelve-month averaging in roughly two-thirds of rolling ten-year periods across all three markets (Shtekhman et al., 2012). Brennan et al. (2005) revisited the question in a continuous-time setting and likewise found no risk-based justification for the practice.

Taken together this strand is remarkably consistent, and its logic is not in dispute. What is in dispute is its scope. Every one of these studies frames the problem as the deployment of an existing windfall: a sum already in hand that is either invested at once or released gradually, with the undeployed balance earning cash returns. That framing is not what a SIP is. An Indian SIP investor does not hold ₹13,20,000 in a bank account and release it slowly; the instalments are funded from monthly income that does not yet exist at the start of the horizon. The comparison of interest to Indian retail investors is therefore between two routes of equal total capital deployed over an identical horizon, which is not the comparison this literature makes. The theoretical results consequently constrain, but do not settle, the question examined here.

3.2 The Behavioural and Measurement-Based Defence

A second strand accepts the expected-return arithmetic but disputes that expected return is the appropriate criterion. Statman (1995) argued that dollar-cost averaging persists because it addresses regret aversion and self-control problems that a mean–variance framework does not represent, and that its value lies in the probability that an investor will actually stay invested. Dubil (2005) reframed lifetime averaging as a risk-reduction device rather than a cost-saving one. Most directly relevant to the present study, Leggio and Lien (2003) showed that the ranking of averaging against immediate investment is itself sensitive to the performance measure applied: strategies that appear inferior on one reward-to-risk ratio can appear superior on another that penalises downside deviation rather than total variability.

This last finding is important and, in the Indian literature, largely unabsorbed. It implies that any single-measure verdict on SIP versus lump-sum is contingent on the measure chosen, and that a defensible comparison must report several conceptually distinct lenses side by side rather than selecting one. It also suggests a testable proposition that this strand itself never tests: that the route which looks better on a total-risk measure need not look better on a benchmark-relative one.

3.3 Indian Empirical Evidence on SIP versus Lump-Sum Investing

Indian studies reach conclusions that are almost uniformly favourable to SIPs, and almost uniformly less rigorous than the international literature. Dwivedi and Bhadouria (2025), working with AMFI and NSE secondary data over a decade, concluded that SIPs suit retail and

risk-averse investors because of rupee-cost averaging, reduced timing risk and the discipline the structure imposes, while acknowledging that lump-sum deployment produces higher absolute returns in a continuously rising market. Parulekar (2025) integrated Modern Portfolio Theory, the efficient market hypothesis and behavioural finance in a systematic review and arrived at a comparable conclusion, that SIPs suit investors requiring consistent risk-adjusted growth while lump-sum suits capital-advantaged and more risk-tolerant investors. Vijiyakumar and Devi (2024) compared the two routes across equity, hybrid and ELSS categories using annualised return, standard deviation, the Sharpe ratio and maximum drawdown, and reported SIPs to be the more consistent approach in volatile conditions.

Three problems recur across this strand. First, most of these studies are descriptive and comparative rather than inferential: differences between routes are reported as point estimates and interpreted as findings, without a test of whether they are distinguishable from sampling noise. Second, several conclusions rest on a comparison of a SIP's money-weighted return against a lump-sum's time-weighted return without acknowledging that the two are not directly commensurable - a problem Mehta and Narzary (2026) identify explicitly, noting that fund performance across the Indian industry is still evaluated through time-weighted, lump-sum measures despite the dominance of SIP investors, and proposing a money-weighted information ratio to close the gap. Third, and most relevant here, funds of different market capitalisations are typically pooled, so that a conclusion about 'equity funds' may in fact be an artefact of the tier composition of the sample.

3.4 Risk-Adjusted Evaluation Frameworks and Their Application in India

The measurement apparatus used in this field descends from a small number of foundational contributions. Markowitz (1952) established the mean–variance representation of portfolio choice from which total-risk measures derive. Sharpe (1966, 1994) proposed excess return per unit of total volatility; Treynor (1965) proposed excess return per unit of systematic risk; and Jensen (1968) introduced the intercept of a market-model regression as a measure of benchmark-relative performance. Sortino and van der Meer (1991) subsequently argued that penalising upside and downside variability symmetrically misrepresents investor preferences, and substituted deviation below a minimum acceptable return.

Indian applications of this apparatus are numerous but narrow. Sawant et al. (2023) evaluated twelve diversified equity funds over 2015–2019 using the Sharpe, Treynor and Jensen measures and found that most did not outperform their benchmarks on either absolute or risk-adjusted terms. Bhagyasree and Kishori (2016) and Virparia (2022) report comparable exercises. The consistent pattern is that these studies apply the measures to raw NAV return series - that is, implicitly to a lump-sum holding - and rarely include the Sortino ratio at all. The apparatus is thus well established in the Indian context, but it has not been turned on the SIP cash-flow structure itself, and the four lenses have not been reported together in a way that would allow the total-risk, downside-risk and benchmark-relative verdicts to be compared against one another.

3.5 Market-Capitalisation Tier as a Conditioning Variable

A fourth strand segments funds by capitalisation but does not compare investment routes. Madhukumar and Ravichandran (2018) compared large-cap and mid-cap fund returns and found performance differences between the categories, and Virparia (2022) reported that large- and small-cap investments were more rewarding over short horizons while mid-cap investments were more rewarding over long ones. Sharma and Tripathi (2023) likewise compared SIP-based performance and risk across small-, mid- and large-cap Indian schemes, reinforcing that risk-adjusted outcomes diverge meaningfully by tier. Such findings establish that tier is a material conditioning variable for fund performance. They do not, however, address whether tier conditions the choice of investment route - which is the relevant question if, as the theoretical strand implies, the value of averaged entry depends on the volatility and cycle behaviour of the segment being entered.

3.6 Research Gap and Its Relevance to the Present Study

Synthesising across these strands, the literature contains a rigorous theoretical argument framed around a scenario that does not correspond to Indian SIP practice; an Indian empirical consensus favouring SIPs that is largely descriptive, frequently pools tiers and seldom tests significance; a measurement literature establishing that the verdict depends on the lens applied, which Indian work has not acted upon; and a tier-segmentation literature that never varies the investment route. The gap that follows is specific: no study appears to have compared SIP and lump-sum investing in Indian equity funds while simultaneously (a) holding market-capitalisation tier

constant, (b) reporting total-risk, downside-risk, systematic-risk and benchmark-relative measures side by side, and (c) testing the resulting differences for statistical significance.

This gap matters because the three omissions interact. Without tier segmentation, one cannot know whether a pooled advantage is uniform or is concentrated in particular segments. Without multiple lenses, one cannot distinguish an advantage in compensation-for-risk from an advantage in benchmark-relative performance. Without significance testing, one cannot know whether either is distinguishable from noise. The present study is designed to address all three simultaneously, and in doing so to establish not merely whether SIP outperforms lump-sum in Indian equity funds, but where it does so and why.

4. Methodology

4.1 Research Design and Approach

The study adopts a quantitative, analytical and comparative research design implemented through historical simulation. It is analytical rather than descriptive because it does not merely report observed fund returns but constructs two counterfactual investor portfolios from the same underlying NAV series and compares their outcomes under controlled conditions. The design is longitudinal, covering 132 consecutive months, and ex post facto, in that it uses realised historical data over which the researcher exercised no control. A simulation design was chosen because the research question concerns two mutually exclusive investment routes that the same investor cannot follow simultaneously in the same fund; only simulation allows both to be observed on identical securities over an identical horizon with identical total capital.

4.2 Research Setting and Context

The setting is the Indian open-ended equity mutual fund market over January 2015 to December 2025. This window was selected deliberately rather than by convenience: it contains four distinct downturn-and-recovery cycles - the 2015–16 correction, the 2018 mid- and small-cap sell-off, the February–April 2020 COVID-19 crash and the 2022 rate-hike sell-off - and therefore exposes both routes to repeated market stress rather than to a single uninterrupted expansion, under which any averaging effect would be mechanically disadvantaged.

4.3 Target Population and Sample

The target population is all open-ended, actively managed Indian equity mutual fund schemes classified under the SEBI large-cap, mid-cap and small-cap categories with a continuous NAV history spanning the full study window. From this population a sample of nine schemes was drawn, comprising one large-cap, one mid-cap and one small-cap scheme from each of three asset management companies: HDFC Asset Management, ICICI Prudential Asset Management and SBI Funds Management. All nine are Direct plans with the Growth option.

The sample size of nine reflects the structure of the research question rather than a target for statistical power in isolation. Because the design is paired - each scheme contributes one SIP observation and one lump-sum observation on the same securities - the effective unit of analysis is the within-scheme difference, which removes fund-specific and manager-specific variation from the comparison and permits meaningful inference from a modest number of schemes. Three schemes per tier is nonetheless a recognised constraint on tier-level inference and is treated as such in the interpretation of results.

4.4 Sampling Technique and Procedure

Purposive (judgemental) sampling was used, applied within a stratified frame. The strata are the three market-capitalisation tiers, which are the study's conditioning variable and are therefore held equal by construction with three schemes in each. Within each stratum, schemes were selected against three criteria: the asset management company must be among the largest in India by assets under management, so that the schemes are economically representative rather than peripheral; the scheme must have a complete daily NAV record across the entire window, so that no observation is imputed; and the same three asset management companies must appear in all three tiers, so that any tier-level difference is not confounded by differences in fund house. Random sampling was rejected because it could not guarantee equal tier representation or complete NAV histories at this sample size, and would have permitted fund-house composition to vary across tiers and confound the tier comparison.

4.5 Data Sources and Collection Method

The study uses only published secondary data, collected through documentary compilation from official sources. Three series were assembled:

- Fund NAV data: daily Net Asset Values for the nine Direct-Growth schemes from January 2015 to December 2025, sourced from the AMFI NAV history repository.
- Benchmark data: daily closing levels of the NIFTY 100 Total Returns Index, the NIFTY Midcap 150 TRI and the NIFTY Smallcap 250 TRI. Total Returns Indices are used in preference to price indices because they incorporate dividends and are therefore the correct comparator for a growth-option scheme.
- Risk-free rate: the weighted average yield on 91-day Government of India Treasury Bills from RBI weekly auction data, averaged to a monthly series and averaging approximately 5.8 per cent per annum across the window.

No primary data were collected and no human participants were involved, so no research instrument such as a questionnaire or interview schedule was required. The analytical tool used was a custom computational workbook (MF_SIP_vs_Lumpsum_Analysis.xlsx), in which all metrics were computed from the three source series.

For reproducibility, the underlying extraction procedure is documented here and reproduced in full in the accompanying workbook, MF_SIP_vs_Lumpsum_Analysis.xlsx, which also contains the raw series, intermediate calculations and every summary table reported in Section 5. Data collection proceeded in five stages:

- AMFI daily NAV data, the core input for returns, volatility and XIRR: obtained from the NAV History section of amfiindia.com for the nine sampled schemes (Direct Plan, Growth Option) over 1 January 2015 to 31 December 2025, downloaded as CSV/Excel; ACE MF and CMIE Prowess were treated as institutional-access alternatives for the same series.
- Benchmark Total Returns Index data, required for Beta, Treynor and Jensen's Alpha: daily and monthly series for the NIFTY 100 TRI, NIFTY Midcap 150 TRI and NIFTY Smallcap 250 TRI were obtained from the Historical Data / Reports section of niftyindices.com for the same window, using Total Returns rather than Price Index values.
- The 91-day Treasury Bill risk-free rate, required for the Sharpe, Sortino, Treynor and Jensen's Alpha calculations: weighted average yields were obtained from the RBI

Database on Indian Economy (DBIE) and the RBI Handbook of Statistics, then converted from an annualised to a monthly rate (yield ÷ 12) to align with the monthly return series.

- Market-capitalisation categorisation and AUM thresholds, used to justify the sample and the tier boundaries: the SEBI categorisation circular of 6 October 2017 and AMFI's latest half-yearly market-capitalisation list were used to confirm the Large- (rank 1–100), Mid- (101–250) and Small-cap (251+) cut-offs, and AMFI's monthly AUM data (as of December 2024) informed selection of the leading schemes within each tier.
- Scheme metadata for descriptive context: inception dates, total expense ratios, stated benchmarks, exit-load structures and fund-manager tenure were drawn from each scheme's Scheme Information Document and the most recent AMC factsheet, and current monthly SIP contribution totals were taken from AMFI's statistics section.

4.6 Investment Simulation Design

For each of the nine schemes, two portfolios were constructed from an identical total capital of ₹13,20,000 over an identical horizon:

- SIP route: ₹10,000 invested on the first trading day of each of the 132 months, with units accumulated at each month's opening NAV.
- Lump-sum route: the entire ₹13,20,000 invested on the first trading day of January 2015 and held without further transactions until December 2025.

Equalising total capital across the two routes is the central control in the design. It ensures that any difference in outcome is attributable to the path of entry rather than to one route committing more money than the other. Both portfolios were marked to market on the final trading day of every month as cumulative units multiplied by the month-end NAV.

4.7 Variables and Measures

The independent variables are the investment route (SIP or lump-sum) and the market-capitalisation tier (large, mid or small). The dependent variables are the six performance and risk measures defined below. All measures are annualised.

4.7.1 Return measures

For the lump-sum route, the annualised return is the compound annual growth rate, CAGR = (Final Value / Initial Investment)^(1/n) − 1, where n is the number of years. For the SIP route, the annualised return is the internal rate of return (XIRR) computed on the actual cash-flow stream of 132 monthly outflows of ₹10,000 and one terminal inflow equal to the December 2025 redemption value. A money-weighted measure is required here because a compound growth rate cannot meaningfully be applied to staggered cash flows; this is also the convention used in AMFI and consolidated account statements.

4.7.2 Monthly return series

Risk measures require a comparable monthly return series for both routes. For the lump-sum route this is the ordinary month-on-month percentage change in portfolio value. For the SIP route, a Modified Dietz return is computed as (V_end − V_begin − CF) / (V_begin + CF), where CF is the ₹10,000 contribution, treated as fully time-weighted because it is invested on the first trading day of the period. This adjustment removes the distorting effect of recurring contributions and yields a series directly comparable to the lump-sum series; it is the series used for standard deviation, downside deviation and the CAPM regression.

4.7.3 Risk and risk-adjusted measures

- Standard deviation: the standard deviation of the monthly return series multiplied by √12, capturing total variability of the investment path.
- Downside deviation: √(mean[min(R_t − MAR, 0)²]) × √12, with the minimum acceptable return set equal to the contemporaneous monthly risk-free rate, so that only sub-target variability is penalised.
- Sharpe ratio: (annualised return − R_f) ÷ annualised standard deviation, measuring excess return per unit of total risk (Sharpe, 1966).
- Sortino ratio: (annualised return − R_f) ÷ downside deviation, measuring excess return per unit of downside risk (Sortino & van der Meer, 1991).
- Treynor ratio: (annualised return − R_f) ÷ Beta, measuring excess return per unit of systematic risk (Treynor, 1965).

- Beta and Jensen's Alpha: estimated by ordinary least squares regression of monthly excess fund returns on monthly excess benchmark returns, $(R_{fund} - R_f) = \alpha + \beta(R_{benchmark} - R_f) + \varepsilon$, with α annualised as twelve times the monthly intercept (Jensen, 1968).

Large-cap schemes are benchmarked against the NIFTY 100 TRI, mid-cap schemes against the NIFTY Midcap 150 TRI and small-cap schemes against the NIFTY Smallcap 250 TRI. Because the Midcap 150 and Smallcap 250 TRI histories in the source data begin in January 2016, Beta, Jensen's Alpha and Sortino estimates for mid- and small-cap schemes rest on 119 monthly observations against 132 for large-cap schemes.

4.8 Data Analysis Techniques

Analysis proceeded in three stages. First, all six measures were computed for each of the eighteen scheme–route combinations and averaged within each tier. Second, a paired difference was calculated for every scheme as the SIP value minus the lump-sum value on each measure, so that the comparison is within-scheme and fund-specific effects cancel. Third, one-sample t-tests were applied to these paired differences under the null hypothesis that the mean difference equals zero, conducted both pooled across all nine schemes ($n = 9$, $df = 8$) and separately within each tier ($n = 3$, $df = 2$). A paired design was selected because the two routes are applied to the same underlying securities and are therefore not independent samples; an independent-samples test would overstate the residual variance and understate the precision of the comparison. Statistical significance is assessed at the conventional 5 per cent level, and the reduced power of the tier-level tests at $n = 3$ is treated explicitly in the interpretation.

4.9 Ethical Considerations

The study involves no human participants, no personal or confidential data and no primary data collection, so questions of informed consent and participant confidentiality do not arise. The ethical obligations that do apply concern the integrity of secondary data use. All data were obtained from publicly available official sources and are reported without selective omission: all nine schemes selected at the outset are reported, including those whose results are less favourable to the study's overall finding, and results that fail to reach statistical significance are reported as such rather than suppressed. The simulation is of hypothetical portfolios and is not

presented as investment advice, and all figures reported are reproducible from the accompanying workbook and its underlying monthly series.

5. Results

This section presents the outcomes of the simulation. Scheme-level results are reported first, followed by tier-level averages, the paired differences between routes and their statistical significance, and finally the drawdown and market-cycle evidence that contextualises those differences.

5.1 Scheme-Level Performance

Table 1 reports all six measures for each of the nine schemes under both routes. The SIP route recorded a higher annualised return than the lump-sum route in every one of the nine schemes, with differences ranging from 0.19 percentage points (SBI Small Cap) to 2.90 percentage points (ICICI Small Cap). Terminal portfolio values are not directly comparable between routes, because lump-sum capital is exposed to the market for the full eleven years while SIP capital is exposed for an average of roughly half that period; the annualised measures are the comparable figures.

Table 1

Scheme-wise annualised performance and risk-adjusted measures, SIP versus lump-sum (2015–2025)

Scheme	Route	Final value (₹)	Ann. return	Std. dev.	Beta	Sharpe	Sortino	Treynor	Jensen's Alpha
HDFC Large Cap	Lump-sum	48,36,759	12.53%	16.88%	1.019	0.396	0.591	0.066	−0.01%
	SIP	31,14,608	14.78%	16.87%	1.019	0.530	0.788	0.088	−0.15%
ICICI Large Cap	Lump-sum	57,72,503	14.36%	15.04%	0.929	0.566	0.838	0.092	2.01%
	SIP	34,13,561	16.31%	15.03%	0.929	0.696	1.027	0.113	1.88%
SBI Large Cap	Lump-sum	52,94,845	13.46%	15.51%	0.950	0.491	0.730	0.080	1.12%
	SIP	30,38,164	14.37%	15.49%	0.951	0.550	0.817	0.090	0.98%
HDFC Mid Cap	Lump-sum	80,92,733	17.93%	18.12%	0.918	0.667	0.979	0.132	1.55%
	SIP	44,04,234	20.52%	18.10%	0.917	0.811	1.188	0.160	1.50%
ICICI Mid Cap	Lump-sum	67,90,952	16.06%	18.92%	0.925	0.540	0.783	0.111	0.17%

Scheme	Route	Final value (₹)	Ann. return	Std. dev.	Beta	Sharpe	Sortino	Treynor	Jensen's Alpha
	SIP	39,74,598	18.83%	18.90%	0.924	0.687	0.994	0.141	0.13%
SBI Mid Cap	Lump-sum	64,10,800	15.45%	18.11%	0.899	0.531	0.750	0.107	−1.48%
	SIP	35,84,924	17.12%	18.08%	0.898	0.624	0.879	0.126	−1.54%
HDFC Small Cap	Lump-sum	79,75,665	17.77%	20.12%	0.844	0.593	0.870	0.141	4.55%
	SIP	41,96,114	19.72%	20.10%	0.843	0.691	1.012	0.165	4.47%
ICICI Small Cap	Lump-sum	63,51,599	15.36%	20.03%	0.823	0.475	0.683	0.116	2.98%
	SIP	38,38,310	18.25%	20.01%	0.822	0.620	0.891	0.151	2.95%
SBI Small Cap	Lump-sum	87,09,732	18.72%	18.79%	0.756	0.685	1.017	0.170	5.02%
	SIP	39,94,024	18.91%	18.77%	0.755	0.696	1.031	0.173	4.96%

Note. Total capital deployed is ₹13,20,000 under both routes. Standard deviation and Jensen's Alpha are annualised. Beta and Jensen's Alpha are estimated against the NIFTY 100 TRI (large-cap), NIFTY Midcap 150 TRI (mid-cap) and NIFTY Smallcap 250 TRI (small-cap). Large-cap estimates use 132 monthly observations; mid- and small-cap estimates use 119. Source: Association of Mutual Funds in India (n.d.) NSE Indices Limited (n.d.) Reserve Bank of India (n.d.) Securities and Exchange Board of India (2017)

5.2 Tier-Level Performance

Averaging across the three schemes in each tier produces the summary reported in Table 2. SIP recorded a higher mean annualised return in all three tiers, exceeding lump-sum by 1.70 percentage points in the large-cap tier, 2.34 percentage points in the mid-cap tier and 1.68 percentage points in the small-cap tier. Mean standard deviation was effectively identical between routes within every tier, differing by no more than 0.03 percentage points, and mean Beta likewise differed by no more than 0.001. The advantage therefore appears on the return side rather than through any reduction in measured risk.

Table 2

Tier-level average performance by investment route

Tier	Route	Ann. return	Std. dev.	Downside dev.	Beta	Sharpe	Sortino	Treynor
Large-cap	Lump-sum	13.45%	15.81%	10.64%	0.966	0.485	0.720	0.079
	SIP	15.15%	15.80%	10.66%	0.966	0.592	0.877	0.097
Mid-cap	Lump-sum	16.48%	18.38%	12.74%	0.914	0.579	0.837	0.116
	SIP	18.82%	18.36%	12.75%	0.913	0.707	1.020	0.142
Small-cap	Lump-sum	17.28%	19.65%	13.43%	0.808	0.584	0.857	0.142

Tier	Route	Ann. return	Std. dev.	Downside dev.	Beta	Sharpe	Sortino	Treynor
	SIP	18.96%	19.63%	13.44%	0.807	0.669	0.978	0.163

Note. Each figure is the unweighted mean of the three schemes in that tier. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

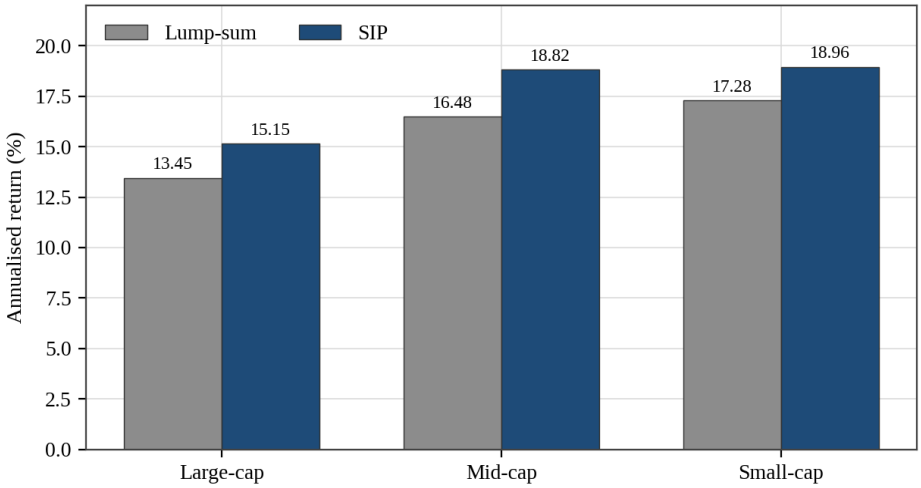


Figure 1. Annualised return by market-capitalisation tier and investment route. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

The return advantage carried through to all three risk-adjusted ratios, as Figure 2 shows. Mean Sharpe ratios rose from 0.485 to 0.592 in the large-cap tier, from 0.579 to 0.707 in the mid-cap tier and from 0.584 to 0.669 in the small-cap tier. Mean Sortino ratios rose from 0.720 to 0.877, from 0.837 to 1.020 and from 0.857 to 0.978 respectively, and mean Treynor ratios from 0.079 to 0.097, from 0.116 to 0.142 and from 0.142 to 0.163. The proportional improvement was largest in the mid-cap tier on every ratio and smallest in the small-cap tier.

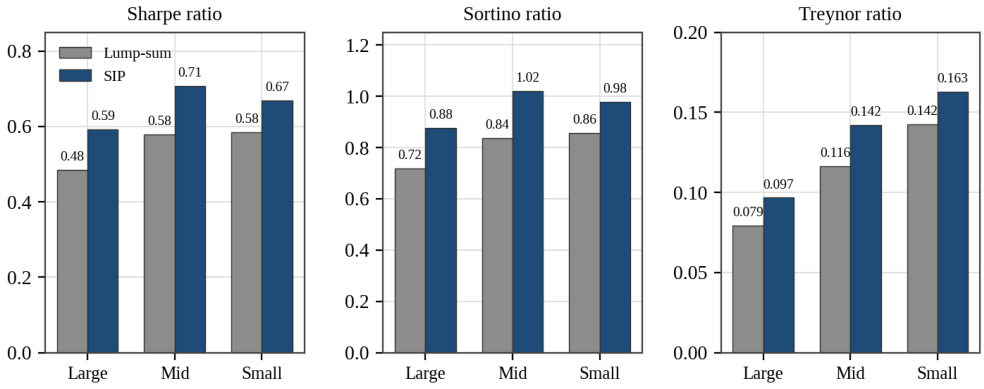


Figure 2. Sharpe, Sortino and Treynor ratios by tier and route.Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

5.3 Paired Differences Between Routes

Table 3 reports the within-scheme difference on each measure, computed as the SIP value minus the lump-sum value. The sign pattern is uniform across all nine schemes: positive on annualised return, Sharpe, Sortino and Treynor, and negative on both standard deviation and Jensen's Alpha. The standard deviation differences are negligible in magnitude, never exceeding 0.03 percentage points, whereas the Jensen's Alpha differences, though small in absolute terms, are consistently negative without exception.

Table 3

Paired differences (SIP minus lump-sum) by scheme

Scheme	Tier	Ann. return	Std. dev.	Sharpe	Sortino	Treynor	Alpha
HDFC Large Cap	Large	+2.25 pp	−0.02 pp	+0.134	+0.197	+0.022	−0.14 pp
ICICI Large Cap	Large	+1.95 pp	−0.00 pp	+0.130	+0.190	+0.021	−0.13 pp
SBI Large Cap	Large	+0.91 pp	−0.01 pp	+0.059	+0.086	+0.010	−0.15 pp
HDFC Mid Cap	Mid	+2.60 pp	−0.02 pp	+0.144	+0.209	+0.029	−0.05 pp
ICICI Mid Cap	Mid	+2.77 pp	−0.01 pp	+0.147	+0.211	+0.030	−0.05 pp
SBI Mid Cap	Mid	+1.67 pp	−0.03 pp	+0.093	+0.130	+0.019	−0.06 pp
HDFC Small Cap	Small	+1.96 pp	−0.03 pp	+0.098	+0.142	+0.023	−0.08 pp
ICICI Small Cap	Small	+2.90 pp	−0.02 pp	+0.145	+0.209	+0.035	−0.03 pp
SBI Small Cap	Small	+0.19 pp	−0.01 pp	+0.011	+0.014	+0.003	−0.06 pp

Note. pp = percentage points. A positive value indicates that SIP exceeded lump-sum on that measure.Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

5.4 Statistical Significance of the SIP–Lump-Sum Differential

Table 4 reports paired-sample t-tests of these differences, conducted pooled across all nine schemes and separately within each tier. Pooled across the sample, the SIP advantage is significant at the 1 per cent level on annualised return, Sharpe, Sortino and Treynor, and the Jensen's Alpha deficit is significant in the opposite direction.

Disaggregated by tier, the pattern is not uniform. On the Sharpe, Sortino and Treynor ratios the SIP advantage is significant at the 5 per cent level in the large-cap tier and more strongly significant in the mid-cap tier, but fails to reach significance in the small-cap tier on any of the three, despite point estimates that continue to favour SIP. On annualised return the large-cap tier

narrowly misses the 5 per cent threshold. The Jensen's Alpha deficit is significant within the large-cap and mid-cap tiers but not within the small-cap tier.

Table 4

Paired-sample t-tests of the SIP–lump-sum differential, pooled and by tier

Measure	Group	n	Mean diff.	t	p	Sig. at 5%
Annualised return	Large-cap	3	+1.70 pp	4.176	.053	No
	Mid-cap	3	+2.34 pp	6.850	.021	Yes
	Small-cap	3	+1.68 pp	2.123	.168	No
	All schemes	9	+1.91 pp	6.446	<.001	Yes
Sharpe ratio	Large-cap	3	+0.108	4.411	.048	Yes
	Mid-cap	3	+0.128	7.335	.018	Yes
	Small-cap	3	+0.085	2.151	.164	No
	All schemes	9	+0.107	6.845	<.001	Yes
Sortino ratio	Large-cap	3	+0.158	4.398	.048	Yes
	Mid-cap	3	+0.183	6.853	.021	Yes
	Small-cap	3	+0.122	2.130	.167	No
	All schemes	9	+0.154	6.772	<.001	Yes
Treynor ratio	Large-cap	3	+0.018	4.357	.049	Yes
	Mid-cap	3	+0.026	7.207	.019	Yes
	Small-cap	3	+0.021	2.153	.164	No
	All schemes	9	+0.021	6.293	<.001	Yes
Jensen's Alpha	Large-cap	3	−0.14 pp	−22.015	.002	Yes
	Mid-cap	3	−0.05 pp	−12.777	.006	Yes
	Small-cap	3	−0.05 pp	−3.439	.075	No
	All schemes	9	−0.08 pp	−5.292	<.001	Yes

Note. One-sample t-tests on within-scheme paired differences under H_0: mean difference = 0. Tier-level tests have df = 2; pooled tests have df = 8. pp = percentage points. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

5.5 Jensen's Alpha and Systematic Risk

Figure 3 presents mean Jensen's Alpha by tier and route. In contrast to every other measure examined, Jensen's Alpha was lower under SIP in all three tiers: 0.90 per cent against 1.04 per cent in the large-cap tier, 0.03 per cent against 0.08 per cent in the mid-cap tier and 4.13 per cent

against 4.18 per cent in the small-cap tier. The absolute size of the deficit is small, never exceeding 0.15 percentage points in any scheme, but its direction is invariant across all nine schemes.

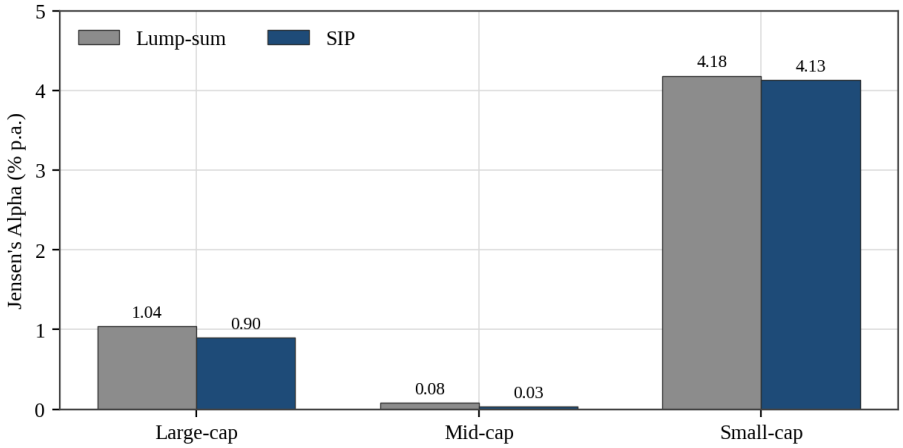


Figure 3. Jensen's Alpha by tier and route. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

Beta estimates were almost indistinguishable between the two routes within every scheme. HDFC Large Cap recorded 1.0194 under lump-sum against 1.0189 under SIP, and SBI Small Cap 0.7561 against 0.7553; no scheme differed by more than 0.0014. The two routes therefore carried essentially identical systematic exposure to their respective benchmarks, and the differences in Sharpe, Sortino and Treynor ratios cannot be attributed to one route bearing more or less market risk than the other.

Figure 4 maps every scheme under both routes in return–risk space. The near-vertical displacement of each SIP point above its lump-sum counterpart is the visual expression of the same result: the SIP route raised return while leaving total risk essentially unchanged. The figure also shows the expected tier ordering, with large-cap schemes clustered at lower volatility and lower return and small-cap schemes at the opposite extreme.

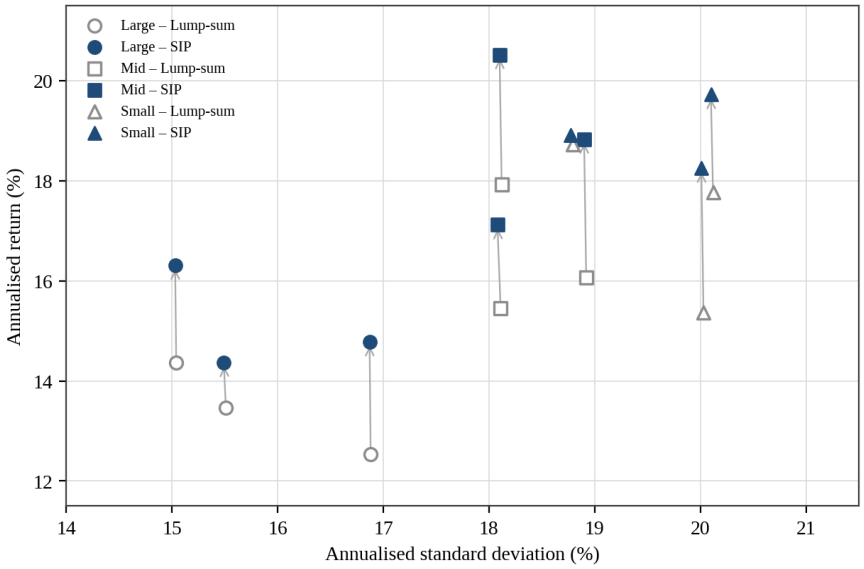


Figure 4. Risk–return map of all nine schemes under both routes. Arrows run from the lump-sum position to the SIP position for the same scheme. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

5.6 Drawdown and Market-Cycle Evidence

Table 5 reports the worst peak-to-trough decline in daily NAV for each scheme over the full window. All nine schemes reached their deepest trough on 23 or 24 March 2020, confirming the COVID-19 crash as the single most severe episode in the period. Drawdowns ranged from −37.08 per cent (SBI Large Cap) to −48.26 per cent (HDFC Small Cap), and were deepest on average in the small-cap tier.

Table 5

Maximum drawdown and trough date by scheme

Scheme	Tier	Maximum drawdown	Trough date
HDFC Large Cap	Large	−40.81%	23 March 2020
ICICI Large Cap	Large	−37.30%	23 March 2020
SBI Large Cap	Large	−37.08%	23 March 2020
HDFC Mid Cap	Mid	−39.51%	23 March 2020
ICICI Mid Cap	Mid	−44.04%	23 March 2020
SBI Mid Cap	Mid	−43.05%	24 March 2020
HDFC Small Cap	Small	−48.26%	24 March 2020
ICICI Small Cap	Small	−45.12%	24 March 2020
SBI Small Cap	Small	−40.26%	24 March 2020

Note. Computed on daily NAV over January 2015 to December 2025 for a single continuous holding. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

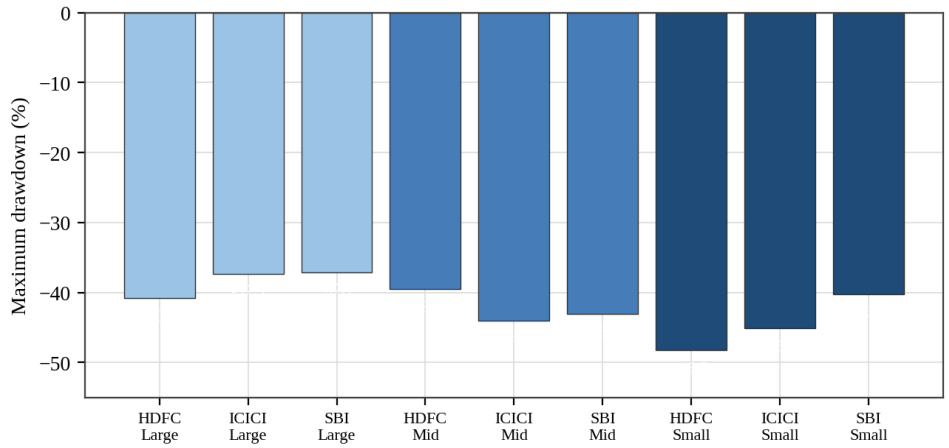


Figure 5. Maximum drawdown by scheme (daily NAV, 2015–2025). Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

Table 6 and Figure 6 set out the benchmark-level declines across the four cycles that the window spans. The tiers were not affected equally. The 2018 sell-off was almost entirely a mid- and small-cap event, costing the NIFTY Midcap 150 TRI 21.77 per cent and the NIFTY Smallcap 250 TRI 33.08 per cent while the NIFTY 100 TRI fell only 4.94 per cent. Recovery times diverged even more sharply: after the 2018 trough the large-cap index regained its prior peak in 213 days, the mid-cap index in 776 days and the small-cap index in 937 days.

Table 6

Benchmark peak-to-trough declines across the four market cycles

Cycle	Benchmark	Decline	Trough	Days to recovery
2015–16 correction	NIFTY 100 TRI	−20.15%	Feb 2016	151
2018 mid/small-cap sell-off	NIFTY 100 TRI	−4.94%	Oct 2018	213
	NIFTY Midcap 150 TRI	−21.77%	Oct 2018	776
	NIFTY Smallcap 250 TRI	−33.08%	Oct 2018	937
COVID-19 crash	NIFTY 100 TRI	−36.95%	Mar 2020	228
	NIFTY Midcap 150 TRI	−34.09%	Mar 2020	245
	NIFTY Smallcap 250 TRI	−39.00%	Mar 2020	405
2022 rate-hike sell-off	NIFTY 100 TRI	−12.96%	Jun 2022	147

Cycle	Benchmark	Decline	Trough	Days to recovery
	NIFTY Midcap 150 TRI	−17.57%	Jun 2022	81
	NIFTY Smallcap 250 TRI	−22.88%	Jun 2022	360

Note. Recovery is the first subsequent date on which the index closed at or above its pre-decline peak. The NIFTY Midcap 150 and Smallcap 250 TRI histories begin in January 2016, so the 2015–16 correction is measured for the NIFTY 100 TRI only; the full table appears in Appendix C. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

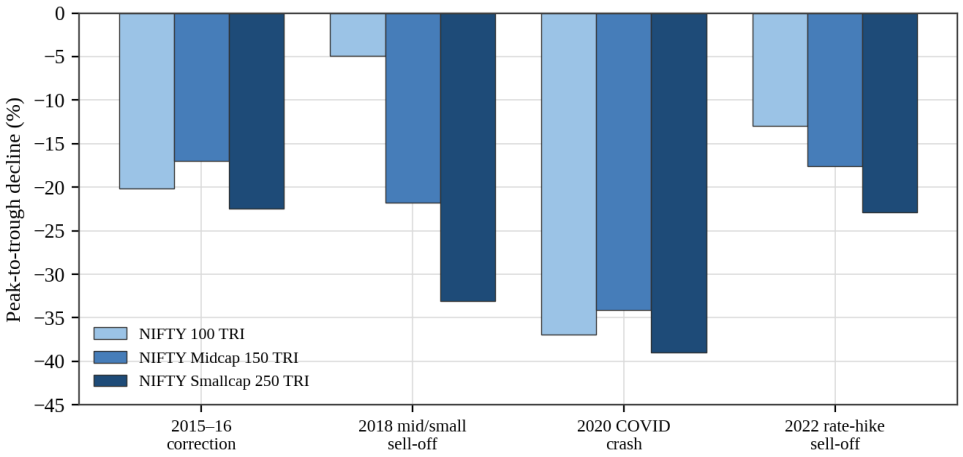


Figure 6. Benchmark declines across the four downturn cycles. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

6. Discussion

6.1 The Scale and Consistency of the SIP Advantage

The first research question asked whether SIP investing delivers higher annualised and risk-adjusted returns than lump-sum investing across the three tiers. The answer from this sample is affirmative and unusually consistent: SIP produced a higher annualised return and a higher Sharpe, Sortino and Treynor ratio in every one of the nine schemes, and the pooled differences are significant well beyond the 1 per cent level. This corroborates the direction of the Indian empirical consensus reported by Dwivedi and Bhadouria (2025), Parulekar (2025) and Vijiyakumar and Devi (2024), and adds what that literature has generally lacked, namely a formal test establishing that the differences are not attributable to sampling variation.

The result sits in apparent tension with the theoretical literature, which holds that staggered entry is suboptimal when the equity risk premium is positive (Constantinides, 1979; Rozeff, 1994; Shtekhman et al., 2012). The tension is largely one of framing rather than of substance. Those studies compare the deployment of a sum already held, with the undeployed balance earning the risk-free rate, so that averaging necessarily sacrifices equity exposure. The present design holds total capital equal but does not hold exposure-time equal, because the SIP instalments are treated as arriving from income rather than from an existing corpus. This is the comparison that Indian retail investors actually face, but it is not the comparison Constantinides addressed, and the two results are therefore compatible rather than contradictory. It also explains why terminal values were substantially larger under lump-sum in every scheme even as annualised returns were lower: the lump-sum corpus was exposed for eleven years, the SIP corpus for roughly half that on average.

6.2 Why the Advantage Narrows in the Small-Cap Tier

The second research question asked whether the advantage is uniform across tiers, and here the disaggregated results are more informative than the pooled ones. The advantage was statistically significant on all three ratios in the large-cap tier, most strongly significant in the mid-cap tier, and significant on none of them in the small-cap tier.

This narrowing is best read as a statistical rather than an economic result. The mean small-cap Sharpe advantage of 0.085 is of the same order as the large-cap advantage of 0.108, and the

mean return advantage of 1.68 percentage points is close to the large-cap figure of 1.70. What differs is the dispersion of those differences across the three small-cap schemes, which ranged from +0.011 to +0.145 on the Sharpe ratio - a spread driven principally by SBI Small Cap, whose lump-sum route performed unusually well and whose SIP advantage was consequently near zero. With only three observations per tier, this dispersion is sufficient to inflate the standard error beyond the point at which a genuine mean difference can be detected. The small-cap tier's higher baseline volatility, at 18.8 to 20.1 per cent annualised against 15.0 to 16.9 per cent for large-cap schemes, is the underlying source of that dispersion.

The mid-cap tier's position as the strongest performer is consistent with the cycle evidence in Table 6. The NIFTY Midcap 150 TRI fell 21.77 per cent in the 2018 sell-off and took 776 days to recover, against 4.94 per cent and 213 days for the NIFTY 100 TRI. Prolonged declines of this kind are precisely the conditions under which a fixed monthly contribution accumulates the most units at depressed prices, and the mid-cap tier experienced a longer such episode than the large-cap tier without the extreme scheme-level dispersion of the small-cap tier.

6.3 Decomposing the Advantage: Timing Rather than Skill

The third research question is the one on which this study's contribution principally rests. Had SIP's superiority reflected genuinely better benchmark-relative performance, Jensen's Alpha should have risen alongside the Sharpe, Sortino and Treynor ratios. It did the opposite. Alpha was lower under SIP in all three tiers and in all nine schemes, and the deficit was statistically significant when pooled and within the large- and mid-cap tiers. Beta, meanwhile, was effectively identical between routes in every scheme, differing by no more than 0.0014, which rules out the possibility that the ratio differences arise from differential market exposure.

The combination is diagnostic. Because Jensen's Alpha measures return in excess of what the Capital Asset Pricing Model attributes to market exposure, and because market exposure was unchanged, a higher Sharpe ratio accompanied by a lower Alpha cannot be produced by superior security selection within the SIP structure. The remaining explanation is the mechanical one. By distributing ₹10,000 purchases across 132 months rather than committing ₹13,20,000 on a single date, the SIP route directed a disproportionate share of its effective capital to the periods that followed market declines, most consequentially the March 2020 trough at which all nine schemes recorded drawdowns of between −37.08 and −48.26 per cent. The units purchased at

those depressed prices raise the money-weighted return without improving the fund's performance relative to its benchmark - which is exactly the signature observed.

This distinction has not, so far as can be established, been drawn cleanly in the tier-segmented Indian literature, which has tended to apply the Sharpe, Treynor and Jensen measures to lump-sum NAV series and report them as a single verdict (Sawant et al., 2023; Virparia, 2022). The finding also vindicates, in a new setting, the more general warning of Leggio and Lien (2003) that the ranking of averaging strategies depends on the performance measure selected. Here the ranking does not merely weaken under an alternative lens; it reverses.

6.4 Implications for Theory and Practice

For the literature, the principal implication is that the familiar conclusion that SIP outperforms lump-sum investing survives decomposition only in a qualified form. It is corroborated on total-risk and downside-risk-adjusted measures, shown to be statistically fragile in the small-cap segment, and mildly reversed on a CAPM-adjusted basis. A pooled, single-lens comparison conceals all three of these qualifications, which is a caution against the design that much of the existing Indian work adopts.

For practitioners, the results support continuing to recommend SIP over lump-sum for large-cap and mid-cap allocations, where the advantage on total- and downside-risk-adjusted measures is both economically meaningful and statistically robust even in a modest sample. For small-cap allocations the recommendation should be tempered: the point estimates still favour SIP, but the advantage was not distinguishable from zero here, which implies that the choice of route in that segment may reasonably be governed by liquidity needs, cash-flow pattern or behavioural preference rather than by an expectation of superior risk-adjusted return. Asset management companies marketing SIP products may likewise wish to calibrate messaging that attributes smoother entry and steadier accumulation to superior fund performance. On this evidence the advantage is real but mechanical, arising from staggered entry through volatile markets rather than from anything the fund or its manager does differently - a claim that is equally valid but analytically distinct.

7. Conclusion

This study set out to close two gaps in the literature comparing SIP and lump-sum investing in Indian equity mutual funds: the absence of segmentation by market-capitalisation tier, and the reliance on a single risk lens. Using 132 months of NAV data for nine Direct-Growth schemes drawn evenly from three tiers and three asset management companies, it simulated a fixed monthly SIP of ₹10,000 against an equivalent lump sum of ₹13,20,000 and evaluated both routes on annualised return, standard deviation, the Sharpe, Sortino and Treynor ratios, and Jensen's Alpha against tier-appropriate Total Returns Indices.

Addressing the first research question, SIP investing produced higher annualised returns and higher Sharpe, Sortino and Treynor ratios than lump-sum investing in every tier and in every one of the nine schemes examined, and these differences were statistically significant when pooled across the sample. On this evidence, the direction of the prevailing conclusion in the Indian literature is confirmed and, for the first time in this design, tested.

Addressing the second research question, the advantage was not uniform across segments. It was statistically significant in the large-cap tier, strongest in the mid-cap tier, and not statistically significant in the small-cap tier on any of the three ratios, despite point estimates that continued to favour SIP. The narrowing reflects the greater dispersion of scheme-level outcomes within the more volatile small-cap tier rather than the disappearance of the underlying advantage.

Addressing the third research question, decomposition through the Capital Asset Pricing Model showed that Jensen's Alpha was lower under SIP in all three tiers while Beta was effectively unchanged. SIP's advantage in this sample is therefore a path-dependent, timing-driven phenomenon attributable to rupee-cost averaging through repeated market declines, and not evidence of improved benchmark-relative performance.

The overall conclusion is that the choice between SIP and lump-sum investing is better framed as a question about the volatility of the segment being entered and the investor's cash-flow position than as a question about which route produces better fund performance. SIP's advantage is genuine, economically meaningful and, in large- and mid-cap allocations, statistically robust - but it is mechanical in origin, and it should be recommended on that basis.

8. Limitations of the Study

Several limitations qualify these conclusions and should be borne in mind when interpreting them.

- Sample size. The sample comprises only three schemes per tier, one from each of three asset management companies. While sufficient to detect the effect in the pooled nine-scheme test, this leaves the tier-level tests underpowered at n = 3 and vulnerable to scheme-specific noise, as the small-cap results illustrate. A broader cross-section of asset management companies would strengthen tier-level inference.
- Data availability. The NIFTY Midcap 150 and NIFTY Smallcap 250 TRI histories in the source data begin in January 2016 rather than January 2015. Beta, Jensen's Alpha and Sortino estimates for mid- and small-cap schemes therefore rest on 119 monthly observations rather than the 132 available for large-cap schemes, and the 2015–16 correction is characterised at the large-cap level only.
- Entry-date sensitivity. The analysis assumes a fixed, unchanging monthly contribution and a single lump-sum entry date of January 2015. Results for both routes are sensitive to the entry date chosen and would differ under alternative starting points, particularly given how much of SIP's advantage traces to its exposure to the March 2020 trough.
- Costs and taxation. The study nets neither exit loads, capital-gains taxation nor differences in total expense ratio between schemes, any of which could narrow or widen the routes' realised after-cost returns in practice.
- Model specification. Jensen's Alpha and Beta were estimated using a single-factor CAPM regression. A multi-factor specification such as the Fama–French–Carhart model might attribute a different share of the return differential to systematic exposures not captured by market Beta alone.
- Scope. The findings apply to actively managed equity schemes of three large asset management companies over one eleven-year window in a single market, and should not be generalised to passive schemes, other asset classes or other markets without further testing.

9. Directions for Future Research

Four extensions follow directly from the findings and limitations above.

- Widening the cross-section beyond three asset management companies per tier, in order to restore statistical power to tier-specific comparisons and to establish whether the small-cap result is a power constraint or a genuine absence of effect.
- Testing sensitivity to the entry date by rolling the eleven-year window across multiple start dates, so as to determine whether the observed SIP advantage is robust to entry-point selection or is substantially an artefact of the March 2020 trough falling within this particular window.
- Re-estimating Jensen's Alpha within a multi-factor asset-pricing framework, in order to isolate size, value and momentum exposures from pure market Beta - a refinement of particular interest given that the small-cap schemes recorded the largest Alphas in this sample.
- Extending the comparison to net-of-cost and post-tax terms, to assess whether the practical gap an investor actually realises between the two routes differs materially from the pre-cost gap documented here.

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11. Appendices

Appendix A: Sample Schemes, Benchmark Mapping and Data Sources

Table A1

Sample schemes and benchmark mapping

Tier	Scheme (Direct–Growth)	AMC	Benchmark index	Observations
Large-cap	HDFC Large Cap Fund	HDFC	NIFTY 100 TRI	132
Large-cap	ICICI Prudential Large Cap Fund	ICICI Prudential	NIFTY 100 TRI	132
Large-cap	SBI Large Cap Fund	SBI	NIFTY 100 TRI	132
Mid-cap	HDFC Mid Cap Fund	HDFC	NIFTY Midcap 150 TRI	119
Mid-cap	ICICI Prudential Mid Cap Fund	ICICI Prudential	NIFTY Midcap 150 TRI	119
Mid-cap	SBI Mid Cap Fund	SBI	NIFTY Midcap 150 TRI	119
Small-cap	HDFC Small Cap Fund	HDFC	NIFTY Smallcap 250 TRI	119
Small-cap	ICICI Prudential Small Cap Fund	ICICI Prudential	NIFTY Smallcap 250 TRI	119
Small-cap	SBI Small Cap Fund	SBI	NIFTY Smallcap 250 TRI	119

Note. Observations refer to the number of monthly return observations available for Beta, Jensen's Alpha and Sortino estimation. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

Data sources used in the study were the AMFI daily NAV repository for scheme NAVs, NSE Indices for the three Total Returns Index series, and Reserve Bank of India weekly auction results for the 91-day Treasury Bill weighted average yield used as the risk-free rate.

Appendix B: Extract from the Monthly Simulation Series

The full monthly series for all nine schemes is contained in Sheet 8 of the accompanying workbook. The extract below illustrates the structure of that series for one scheme, HDFC Large Cap, showing the first twelve months and the final six months of the simulation.

Table B1

Extract of the monthly simulation series - HDFC Large Cap

Month	NAV (start)	NAV (end)	Lump-sum value (₹)	Lump-sum return	SIP value (₹)
Jan-2015	349.31	364.72	13,78,213	+4.41%	10,441

Month	NAV (start)	NAV (end)	Lump-sum value (₹)	Lump-sum return	SIP value (₹)
Feb-2015	364.14	359.30	13,57,724	−1.49%	20,153
Mar-2015	366.01	347.13	13,11,758	−3.39%	28,955
Apr-2015	350.75	339.04	12,81,168	−2.33%	37,945
May-2015	344.67	346.35	13,08,822	+2.16%	48,813
Jun-2015	347.33	341.89	12,91,957	−1.29%	58,028
Jul-2015	346.71	354.62	13,40,039	+3.72%	70,416
Aug-2015	357.54	330.63	12,49,419	−6.76%	74,901
Sep-2015	322.10	328.57	12,41,604	−0.63%	84,633
Oct-2015	327.79	333.90	12,61,775	+1.62%	96,195
Nov-2015	334.06	336.39	12,71,162	+0.74%	1,06,980
Dec-2015	336.59	329.75	12,46,086	−1.97%	1,14,667
⋮	⋮	⋮	⋮	⋮	⋮
Jul-2025	1,246.47	1,222.70	46,20,419	−2.02%	29,25,961
Aug-2025	1,213.29	1,204.70	45,52,392	−1.47%	28,92,811
Sep-2025	1,215.47	1,216.06	45,95,324	+0.94%	29,30,096
Oct-2025	1,227.38	1,256.68	47,48,817	+3.34%	30,38,206
Nov-2025	1,262.31	1,282.04	48,44,649	+2.02%	31,09,674
Dec-2025	1,280.14	1,279.95	48,36,759	−0.16%	31,14,608

Note. Lump-sum value assumes ₹13,20,000 invested on the first trading day of January 2015. SIP value reflects cumulative units purchased through monthly ₹10,000 contributions valued at month-end NAV. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).

Appendix C: Full Market-Cycle Table

Table C1

Benchmark declines and recovery periods across the four market cycles

Cycle	Benchmark	Decline	Trough date	Recovery date	Days
2015–16 correction	NIFTY 100 TRI	−20.15%	25 Feb 2016	25 Jul 2016	151
	NIFTY Midcap 150 TRI	−16.96%	26 Feb 2016	28 Jun 2016	123
	NIFTY Smallcap 250 TRI	−22.49%	29 Feb 2016	01 Jul 2016	123
2018 sell-off	NIFTY 100 TRI	−4.94%	26 Oct 2018	27 May 2019	213
	NIFTY Midcap 150 TRI	−21.77%	09 Oct 2018	23 Nov 2020	776
	NIFTY Smallcap 250 TRI	−33.08%	09 Oct 2018	03 May 2021	937
COVID-19 crash	NIFTY 100 TRI	−36.95%	23 Mar 2020	06 Nov 2020	228

Cycle	Benchmark	Decline	Trough date	Recovery date	Days
	NIFTY Midcap 150 TRI	−34.09%	23 Mar 2020	23 Nov 2020	245
	NIFTY Smallcap 250 TRI	−39.00%	24 Mar 2020	03 May 2021	405
2022 rate-hike sell-off	NIFTY 100 TRI	−12.96%	17 Jun 2022	11 Nov 2022	147
	NIFTY Midcap 150 TRI	−17.57%	20 Jun 2022	09 Sep 2022	81
	NIFTY Smallcap 250 TRI	−22.88%	20 Jun 2022	15 Jun 2023	360

Note. The 2015–16 decline for the Midcap 150 and Smallcap 250 indices is measured from January 2016, the start of their TRI histories in the source data, and therefore understates the full extent of that correction for those segments. Source: Computed by author using Microsoft Excel; raw data sourced from AMFI (n.d.), NSE Indices Limited (n.d.), and RBI (n.d.).